W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

Pentester Name (Cybersecurity Professional)	Naman Sharma
Contact Information	myarcade.era@gmail.com
Program / Batch	Batch B084 — Networkwalks Cybersecurity Internship
Practical Execution Date	11 November 2026
Report Submission Date	12 November 2026
Modules Completed	W2-PM1 (Elective): Footprinting with Multiple Kali Tools W2-PM5 (Essential): Zenmap-Based Network Scanning W2-PM-FINAL (Essential): Detailed Practical Report
Client / Target	1. `networkwalks.com` (Public-domain footprinting) 2. `10.45.149.0/24` (Personal mobile hotspot local subnet)
Permission Secured from Client?	Yes — Strictly within assigned Week 2 Networkwalks internship practical scope
Phases Covered	Phase 1: Reconnaissance & Footprinting (W2-PM1) Phase 2: Scanning & Network Discovery (W2-PM5)

1. Liability Disclaimer & Ethical Scope Statement

All activities documented in this report were performed strictly within the assigned Week 2 internship practical scope against the designated public domain (`networkwalks.com`) and the intern's own personal mobile hotspot subnet (`10.45.149.0/24`). The completed work consisted exclusively of passive/diagnostic footprinting and local host discovery; no vulnerability exploitation, intrusive probing, or unauthorized access was attempted.

All materials in this report are compiled strictly for educational and research purposes under the Networkwalks Cybersecurity Internship (Batch B084). Observed software versions, response headers, and DNS configuration details are documented as factual technical observations rather than confirmed vulnerabilities. Unauthorized access or testing against systems without explicit authorization is strictly prohibited.

2. Introduction

This report fulfills deliverable **W2-PM-FINAL** for Week 2 of the Networkwalks Cybersecurity Internship (Batch B084). It covers public-domain footprinting of `networkwalks.com` using multiple Kali Linux tools (**W2-PM1**) and local network host discovery on the personal mobile hotspot subnet `10.45.149.0/24` using Zenmap (**W2-PM5**). Other elective modules (W2-PM2 GHDB, W2-PM3 Maltego, and W2-PM4 theHarvester) were not selected for this submission and are not claimed as completed.

All footprinting commands were executed in Kali Linux (`kali-linux-2025.2-virtualbox-amd64`), and local host discovery was performed on a Windows 11 host running Zenmap (Nmap v7.98 / Npcap v1.83). Every step below documents the exact commands executed, verified terminal outputs, captured screenshot evidence, and analytical significance.

3. Written Permission & Authorization to Test

FORMAL AUTHORIZATION & RULES OF ENGAGEMENT

Written permission and academic authorization are confirmed for **Naman Sharma** (Batch **B084**) to perform passive/diagnostic domain footprinting and local network host discovery strictly within the assigned Week 2 scope of the **Networkwalks Cybersecurity Internship**.

- **Authorized External Domain Target:** `networkwalks.com` (non-intrusive WHOIS, DNS, HTTP header, technology fingerprinting, and WAF signature detection only).
- **Authorized Local Network Target:** `10.45.149.0/24` (intern-owned personal mobile hotspot subnet; non-intrusive ICMP/ARP ping scan host discovery via Zenmap only).
- **Explicit Prohibitions:** Active port scanning against external public servers, vulnerability exploitation, denial-of-service testing, unauthorized traffic interception, and probing outside the authorized targets listed above are strictly prohibited.

Role	Name / Authority	Authorization Status	Date
Program Authority	Networkwalks Cybersecurity Internship (Batch B084)	Authorized within Week 2 Practical Scope	11 November 2026
Intern / Performer	Naman Sharma (`myarcade.era@gmail.com`)	Acknowledged & Executed Within Scope	11 November 2026

4. Tools Used

The table below lists each verified tool, confirmed version, operating platform, and purpose in the practical work.

Tool / Component	Verified Version	Platform	Purpose in Practical Work
`WHOIS`	v5.6.6	Kali Linux	Queries public registry and registrar domain registration records (dates, status locks, privacy masking, nameservers).
`WhatWeb`	v0.6.4	Kali Linux	Fingerprints web server software, CMS versions, plugins, frontend libraries, and IP details.
`Nslookup`	DNS lookup utility	Kali Linux	Queries A, NS, and MX records through the local gateway DNS resolver (`10.45.149.194#53`).
`cURL`	HTTP client	Kali Linux	Fetches HTTP response headers (`-I` and `-L -I`) to inspect status lines, caching headers, cookies, and link headers.
`WAFW00F`	v2.4.2	Kali Linux	Sends diagnostic HTTP requests to detect and fingerprint Web Application Firewalls.
`DNSRecon`	v1.3.1	Kali Linux	Performs general DNS enumeration (SOA, NS, BIND version string, MX, A, TXT/SPF, SRV, and DNSSEC query).
`Zenmap / Nmap`	Nmap v7.98 (Npcap v1.83)	Windows 11	Executes a ping scan (`-sn`) for local subnet host discovery and generates a visual network topology.

5. Activities Performed

5.1 Footprinting & Reconnaissance (Module W2-PM1 — Target: networkwalks.com)

Reconnaissance was executed against `networkwalks.com` in Kali Linux using six command-line utilities to collect publicly available registration, web technology, DNS, HTTP header, and WAF metadata:

1. **Domain Registration Query (`whois v5.6.6`):** Executed `whois networkwalks.com`. Identified Registry Domain ID `2452319255_DOMAIN_COM-VRSN`, registrar `GoDaddy.com, LLC` (IANA ID: 146; WHOIS server: `whois.godaddy.com`), creation date `2019-11-06`, updated date `2025-11-12`, expiry date `2027-11-06`, EPP client locks (`clientDeleteProhibited`, `clientRenewProhibited`, `clientTransferProhibited`, `clientUpdateProhibited`), privacy masking via `Registration Private / Domains By Proxy, LLC` (Tempe, Arizona, US), and `DNSSEC: unsigned`. Registry WHOIS listed two nameservers (`NS6135.HOSTGATOR.COM`, `NS6136.HOSTGATOR.COM`), whereas registrar WHOIS listed four (`NS6135.HOSTGATOR.COM`, `NS6136.HOSTGATOR.COM`, `NS29.DOMAINCONTROL.COM`, `NS30.DOMAINCONTROL.COM`).
2. **Web Technology Fingerprinting (`whatweb v0.6.4`):** Executed `whatweb https://networkwalks.com`. Returned `[200 OK]` from IPv4 `192.232.216.135` (geolocation reported as `UNITED STATES [US]`), identifying `HTTPServer[Apache]`, `WordPress[7.1.4]` (via MetaGenerator), `WordPress Download Manager 3.3.76`, `Bootstrap[7.1.4]`, `JQuery[3.7.1]`, `HTML5`, `Open-Graph-Protocol[website]`, `Google-Tag-Manager`, title `Networkwalks Academy`, contact email `info@networkwalks.com`, and `HttpOnly[_wpdm_client]`.

3. **DNS Record Resolution (`nslookup`):** Executed `nslookup networkwalks.com`, `nslookup -type=NS networkwalks.com`, and `nslookup -type=MX networkwalks.com` via local gateway resolver `10.45.149.194#53`. Returned non-authoritative answers resolving `networkwalks.com` to `192.232.216.135`, NS records to `ns6135.hostgator.com` and `ns6136.hostgator.com`, and MX record `0 mail.networkwalks.com` (also resolving to `192.232.216.135`).
4. **HTTP Response Header Inspection (`curl`):** Executed `curl -I https://networkwalks.com` and `curl -L -I https://networkwalks.com`. Both returned `HTTP/2 200` with `server: Apache` and no redirect chain, exposing caching headers (`x-endurance-cache-level: 0`, `x-nginx-cache: WordPress`), `referrer-policy: no-referrer-when-downgrade`, `permissions-policy`, cookie `_wpdm_client` (`path=/; domain=networkwalks.com; secure; HttpOnly`), and `link` headers exposing `https://networkwalks.com/wp-json/` and `https://networkwalks.com/wp-json/wp/v2/pages/53`.
5. **Web Application Firewall Detection (`wafw00f v2.4.2`):** Executed `wafw00f https://networkwalks.com`. Detected `ModSecurity (SpiderLabs) WAF` protecting the site after sending 2 diagnostic HTTP requests.
6. **DNS Record Enumeration (`dnsrecon v1.3.1`):** Executed `dnsrecon -d networkwalks.com`. Enumerated 8 records including SOA (`ns6135.hostgator.com` at `50.87.144.87`), NS (`ns6135.hostgator.com` at `50.87.144.87` and `ns6136.hostgator.com` at `192.232.216.131`), BIND version string `"9.16.23-RH"` on both nameservers, MX (`mail.networkwalks.com` at `192.232.216.135`), A (`192.232.216.135`), TXT records (Google site-verification token and SPF `v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all`), 8 `_autodiscover._tcp.networkwalks.com` SRV records pointing to `cpanelemaildiscovery.cpanel.net` on port 443, and `ERROR No answer for DNSSEC query`.

5.2 Network Scanning with Zenmap (Module W2-PM5 — Subnet: 10.45.149.0/24)

On the Windows 11 host connected to a personal mobile hotspot (subnet `10.45.149.0/24`, subnet mask `255.255.255.0`), Zenmap (Nmap v7.98 / Npcap v1.83) was used with the **Ping scan** profile (`nmap -sn 10.45.149.0/24`) to discover active hosts across all 256 IPv4 addresses without performing port scans.

After connecting a second mobile phone to the hotspot, the scan completed in **8.34 seconds** and detected **3 live hosts**:

Device Identification	IPv4 Address	Reported MAC Address	Observed Status & Latency
Windows Laptop (Scanner)	`10.45.149.25`	Not reported in Zenmap output	Host is up (local scanning interface; MAC not displayed in Zenmap output)
Second Connected Phone	`10.45.149.103`	`42:60:**:**:**:**` (Unknown)	Host is up (`0.023s` latency)

Device Identification	IPv4 Address	Reported MAC Address	Observed Status & Latency
Mobile Hotspot Gateway	`10.45.149.194`	`7E:**:**:**:**:**` (Unknown)	Host is up (`0.0030s` latency); verified in Nmap Output & Host Details tabs

Following completion of the ping sweep, the Zenmap **Topology** tab was opened with the **Topology Legend** enabled, displaying `localhost` linked via dashed lines (no traceroute hop information) to the three green host nodes (`10.45.149.25` , `10.45.149.103` , and `10.45.149.194`), and exported to `W2-PM5_Network_Topology.pdf` .

6. Risk Analysis / Technical Observations & Impact

Because the completed practicals involved passive/diagnostic footprinting and local ping-sweep host discovery only, no vulnerability scanning or exploitation was performed. The observations below distinguish verified tool evidence from potential security implications and testing boundaries.

ID	Observation / Finding	Verified Evidence	Potential Impact & Scope Boundary	Classification
`OBS-01`	WordPress & Plugin Version Disclosure	`WhatWeb v0.6.4` reported `WordPress 7.1.4` and `WordPress Download Manager 3.3.76` via HTML MetaGenerator tags.	Public version strings assist external fingerprinting if components become outdated. No vulnerability testing or exploit validation was performed.	**Medium (Info Disclosure)**
`OBS-02`	DNS BIND Version Disclosure	`DNSRecon v1.3.1` reported BIND version string `"9.16.23-RH"` on nameservers `50.87.144.87` and `192.232.216.131` .	Exposing DNS software version strings reveals underlying nameserver details. No DNS exploitability or zone-transfer (AXFR) testing was conducted.	**Medium (Info Disclosure)**
`OBS-03`	Unobserved HTTP Security Headers	`cURL` did not observe `Strict-Transport-Security` , `Content-Security-Policy` , `X-Frame-Options` , or `X-Content-Type-Options` on root HTTPS response.	Missing defensive headers may reduce browser-side hardening. Only the root URL response was captured; other endpoints were not tested.	**Low (Header Config)**

ID	Observation / Finding	Verified Evidence	Potential Impact & Scope Boundary	Classification
`OBS-04`	WordPress REST API Discovery Link	`cURL` observed HTTP `link` header advertising `https://networkwalks.com/wp-json/` and `/wp-json/wp/v2/pages/53`.	Exposes standard REST API discovery metadata. No requests were sent to REST API routes, so user enumeration was not tested.	**Low (Endpoint Exposure)**
`OBS-05`	Unsigned DNSSEC Status	`WHOIS` reported `DNSSEC: unsigned` and `DNSRecon` received no answer for its DNSSEC query.	Unsigned zones lack cryptographic authentication of DNS responses. No DNS spoofing or cache poisoning was tested.	**Low (DNS Config)**
`OBS-06`	ModSecurity WAF Detection	`WAFW00F v2.4.2` detected `ModSecurity (SpiderLabs) WAF` on `https://networkwalks.com` after 2 diagnostic requests.	Observed defensive control signature. Active blocking rules, logging, and bypass resistance were not tested.	**Informational (Control)**
`OBS-07`	Active Local Subnet Hosts Discovered	`Zenmap (nmap -sn 10.45.149.0/24)` detected 3 live hosts (`10.45.149.25` , `10.45.149.103` , `10.45.149.194`) in 8.34s.	Records active IP/MAC responses on the personal hotspot subnet; does not establish whether wireless client isolation is disabled.	**Low (Host Discovery)**

7. Recommendations

The following conditional security recommendations follow directly from the technical observations recorded in Section 6 and standard operational hygiene:

1. **Review Server and Software Version Disclosure Settings:** Review web server, CMS metadata, and DNS server version-reporting settings (Apache banners, WordPress generator tags, and BIND `version` directives) to suppress unnecessary version strings where appropriate.
2. **Maintain Supported and Updated Software:** Keep Apache, WordPress core, installed plugins (including WordPress Download Manager), and DNS server software on supported releases with timely security updates.
3. **Review HTTP Security-Header Configuration:** Review web server and reverse-proxy header configurations and apply appropriate HTTP security headers (`Strict-Transport-Security` , `Content-Security-Policy` , `X-Frame-Options` , and `X-Content-Type-Options`) where compatible with application functionality.
4. **Review Access Controls on Sensitive WordPress REST API Endpoints:** Review authentication and permission controls on sensitive WordPress REST API routes (such as user enumeration endpoints) rather than disabling the public REST API indiscriminately.
5. **Evaluate DNSSEC Signing:** Evaluate enabling DNSSEC signing with the domain registrar and DNS hosting provider to provide cryptographic authenticity for DNS responses.
6. **Maintain and Monitor WAF Configuration:** Keep ModSecurity (SpiderLabs) WAF active and properly tuned with ongoing log monitoring.
7. **General Wireless Network Hardening:** Treat wireless client isolation (AP isolation) as an optional general network-hardening measure on shared Wi-Fi or personal hotspot networks, and maintain periodic host inventory checks.

8. Conclusion

During Week 2 of the Networkwalks Cybersecurity Internship (Batch B084), modules **W2-PM1**, **W2-PM5**, and **W2-PM-FINAL** were completed strictly within the assigned practical scope. Using six Kali Linux tools (`whois` , `whatweb` , `nslookup` , `curl` , `wafw00f` , and `dnsrecon`), an evidence-backed external profile of `networkwalks.com` was constructed covering registration records, hosting nameservers, web server and WordPress stack metadata, HTTP response headers, ModSecurity WAF detection, and DNS/SRV records.

In the local network scanning phase, Zenmap (Nmap v7.98 / Npcap v1.83) was used to execute a non-intrusive ping sweep (`nmap -sn 10.45.149.0/24`) across the personal mobile hotspot subnet, identifying three active hosts (`10.45.149.25` , `10.45.149.103` , and `10.45.149.194`), recording available MAC addresses, and exporting the visual network topology (`W2-PM5_Network_Topology.pdf`). Throughout both phases, strict analytical restraint was maintained by separating verified tool outputs from unverified assumptions.

9. Evidences Collected (Captured Practical Screenshots)

Below are the original screenshots captured during execution of modules W2-PM1 and W2-PM5 on 11 November 2026:

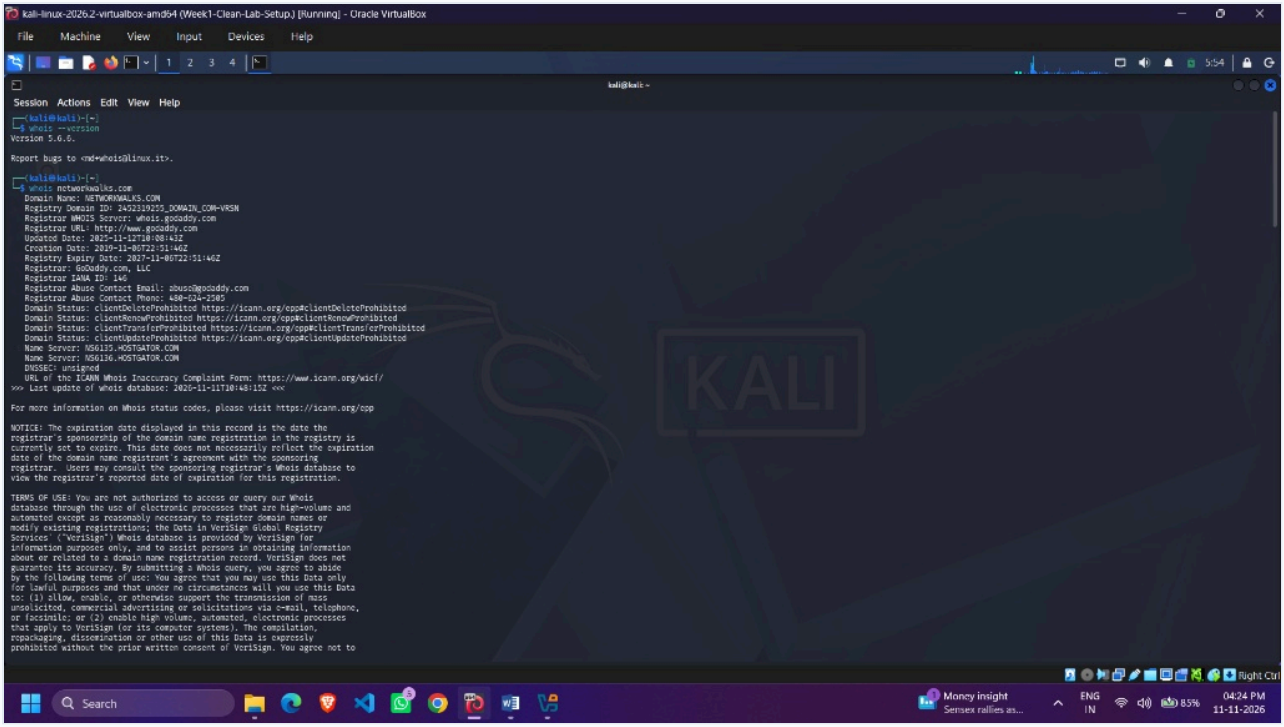


Figure 1: WHOIS Output — Version and Registry Record. Execution of `whois --version` (v5.6.6) and `whois networkwalks.com` showing creation/expiry dates, GoDaddy registrar metadata, EPP client locks, and registry nameservers `NS6135.HOSTGATOR.COM` / `NS6136.HOSTGATOR.COM`.

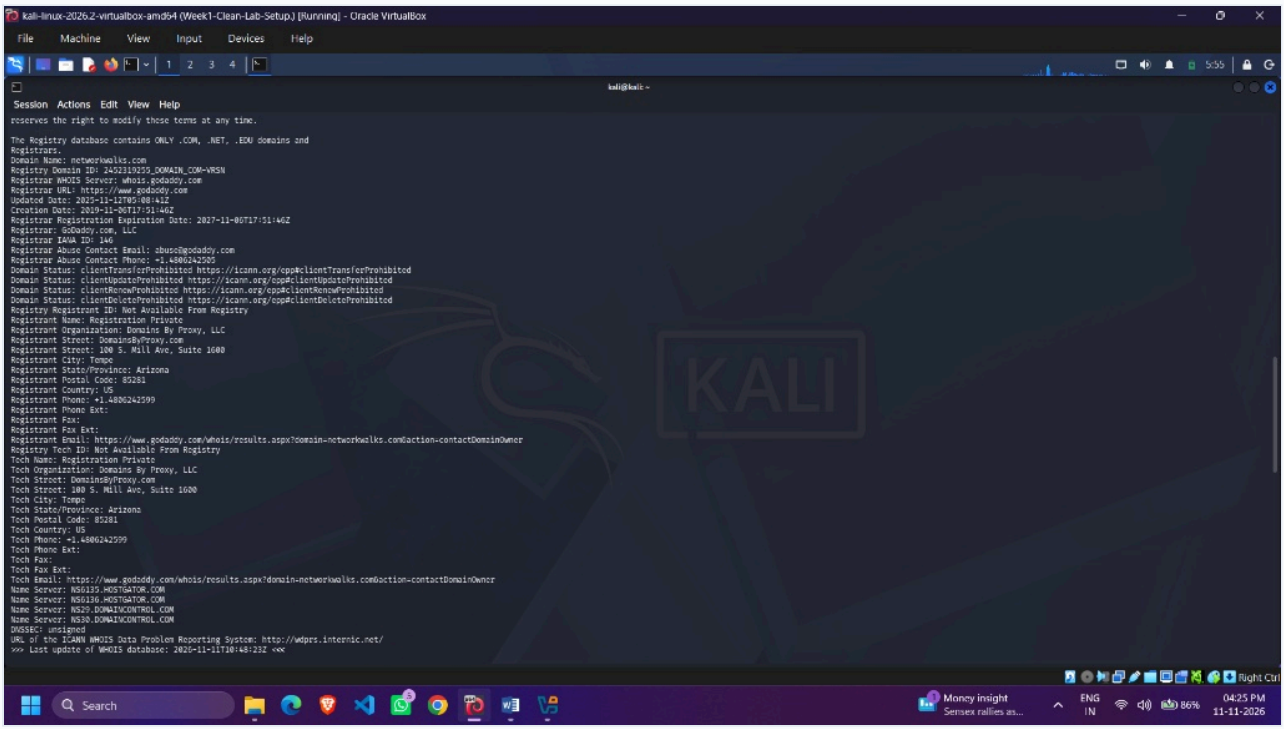


Figure 2: WHOIS Output — Registrar Record and Privacy Masking. Registrar WHOIS output displaying Domains By Proxy, LLC contact masking and four listed nameservers (HostGator and Domain Control).

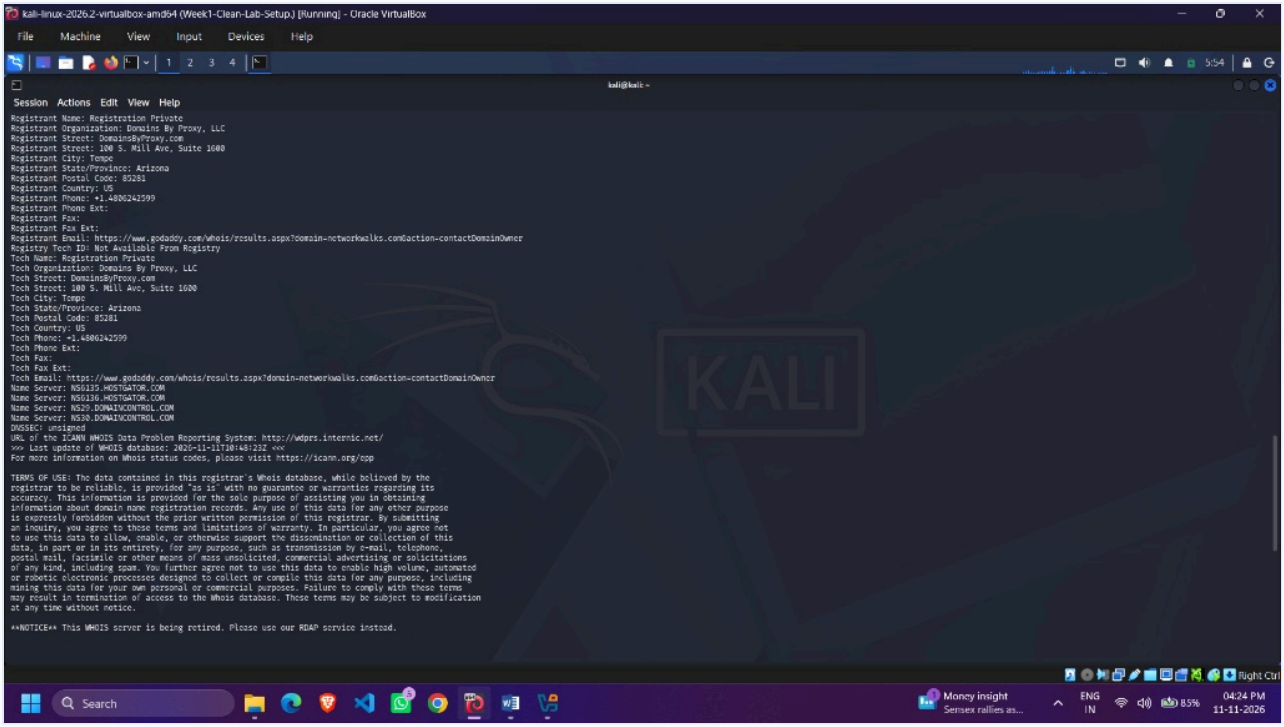


Figure 3: WHOIS Output — Technical Contact Masking and Terms Notice. Continuation of the registrar WHOIS response showing technical privacy fields, `DNSSEC: unsigned`, and registry/registrar notices.

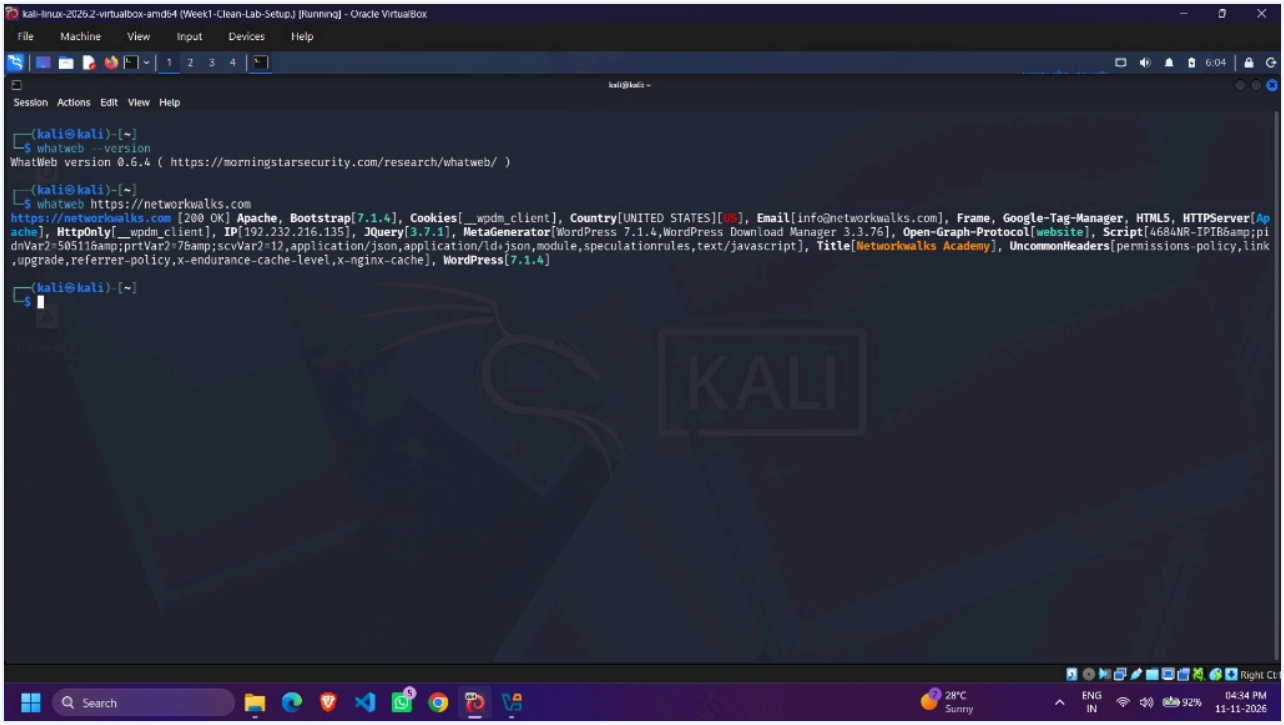


Figure 4: WhatWeb Technology Fingerprinting Output. Output from `whatweb v0.6.4` identifying Apache, WordPress 7.1.4, WordPress Download Manager 3.3.76, Bootstrap 7.1.4, jQuery 3.7.1, and IP `192.232.216.135`.

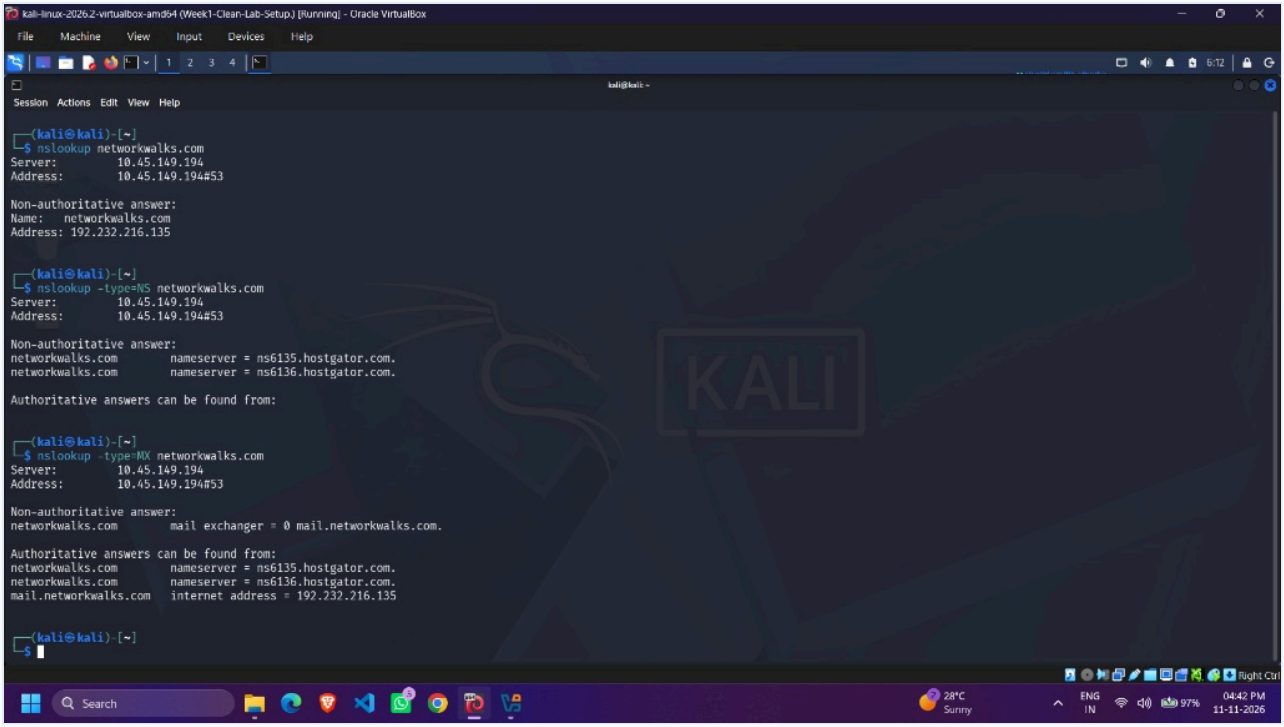


Figure 5: Nslookup DNS Queries (A, NS, and MX). Non-authoritative resolution outputs showing `networkwalks.com` and `mail.networkwalks.com` resolving to `192.232.216.135` and NS records pointing to HostGator.

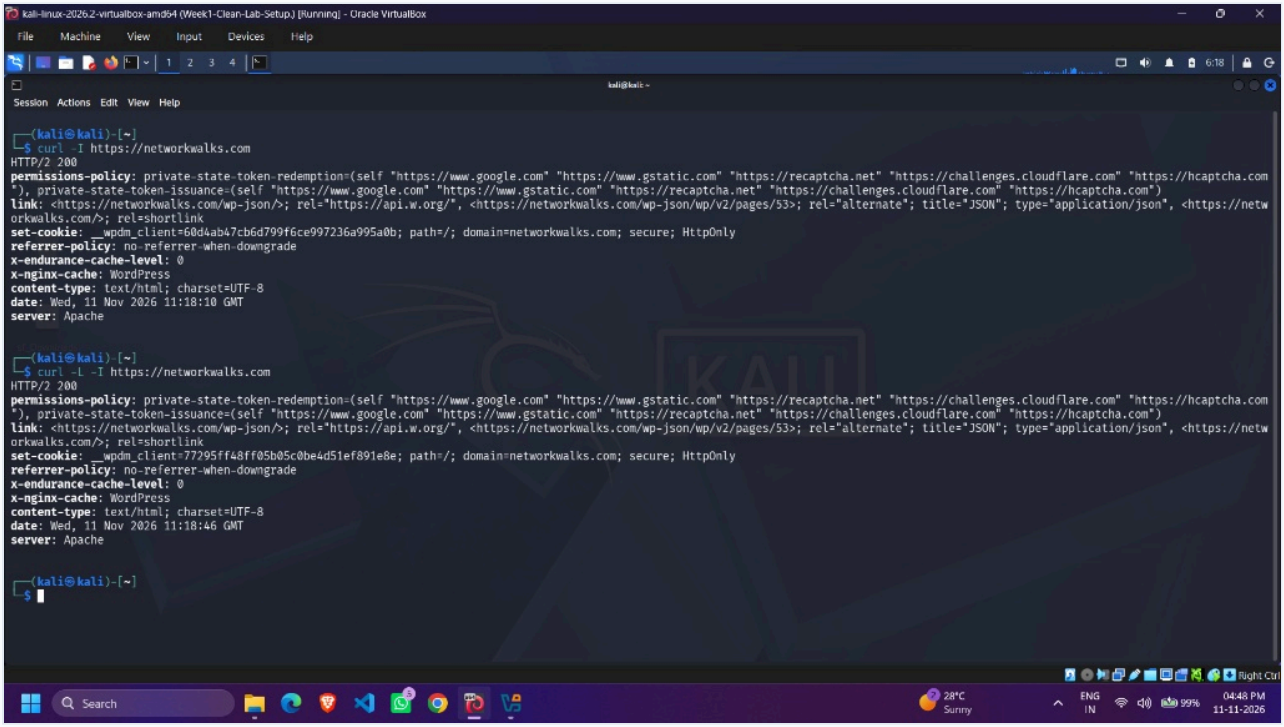


Figure 6: cURL HTTP Response Header Output. Outputs of `curl -I` and `curl -L -I` showing `HTTP/2 200`, `server: Apache`, caching headers, `_wpdm_client` response cookie attributes, and the `/wp-json/` link header.

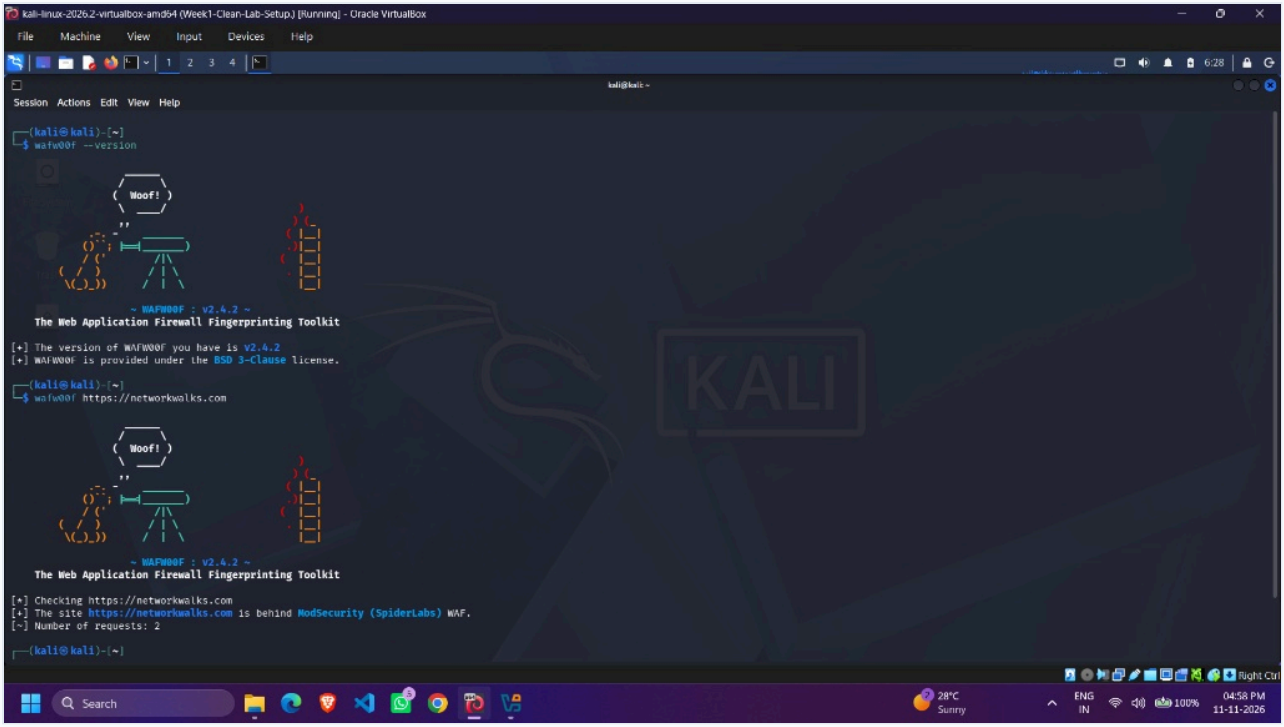


Figure 7: WAFW00F Detection Output. Output from `wafw00f v2.4.2` identifying ModSecurity (SpiderLabs) WAF on `https://networkwalks.com` in 2 requests.

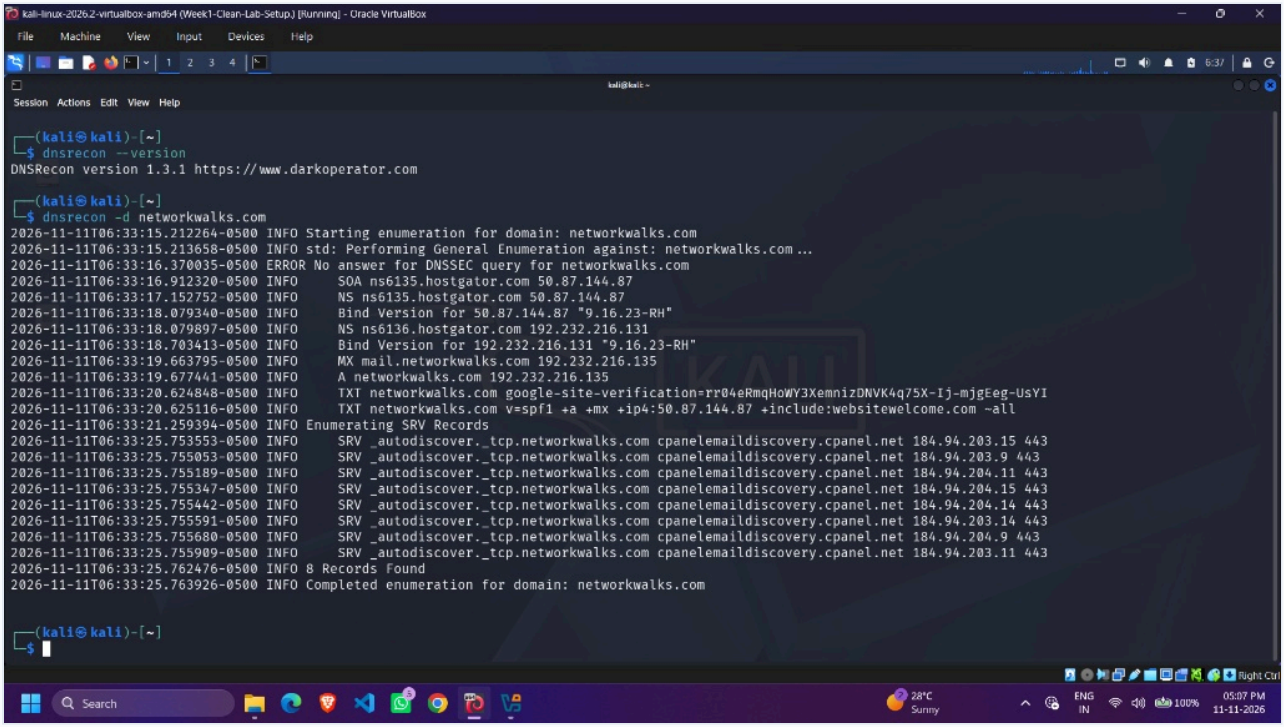


Figure 8: DNSRecon General Enumeration Output. Output from `dnsrecon v1.3.1` reporting SOA, NS, BIND version `"9.16.23-RH"`, MX, A, TXT/SPF, and 8 cPanel autodiscover SRV records.

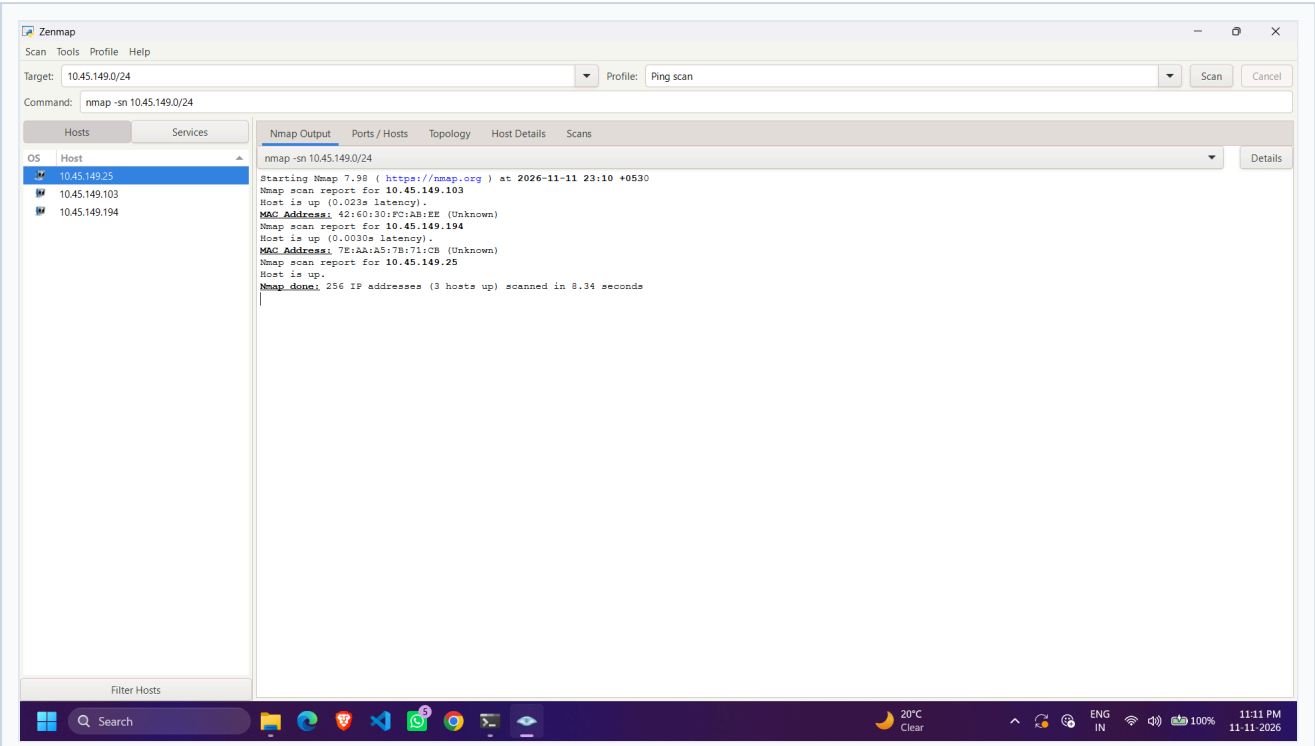


Figure 9: Zenmap Ping Scan Output (10.45.149.0/24). Zenmap Nmap Output tab showing `nmap -sn 10.45.149.0/24` discovering 3 live hosts (`10.45.149.103` , `10.45.149.194` , and `10.45.149.25`) in 8.34 seconds.

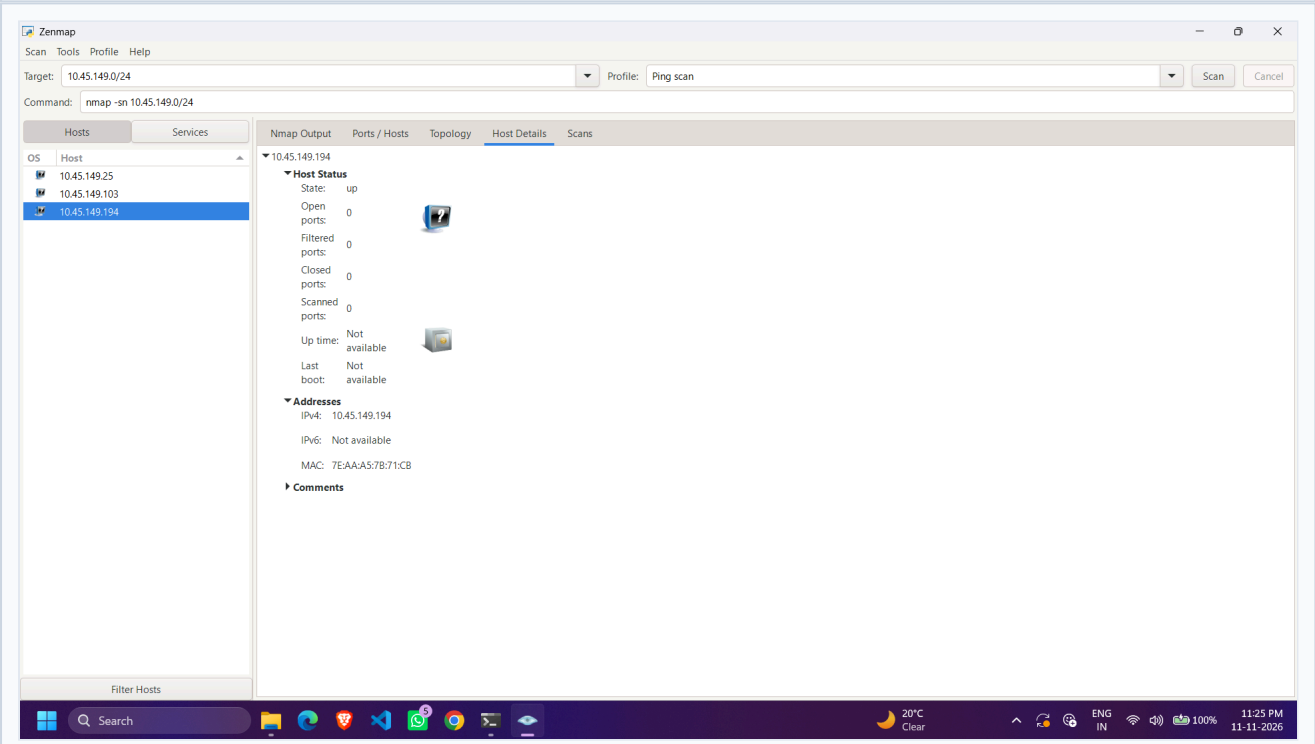


Figure 10: Zenmap Host Details — Gateway MAC Verification. Zenmap Host Details tab for gateway `10.45.149.194` confirming Host Status up, 0 scanned ports, and MAC address `7E:AA:A5:7B:71:CB` .

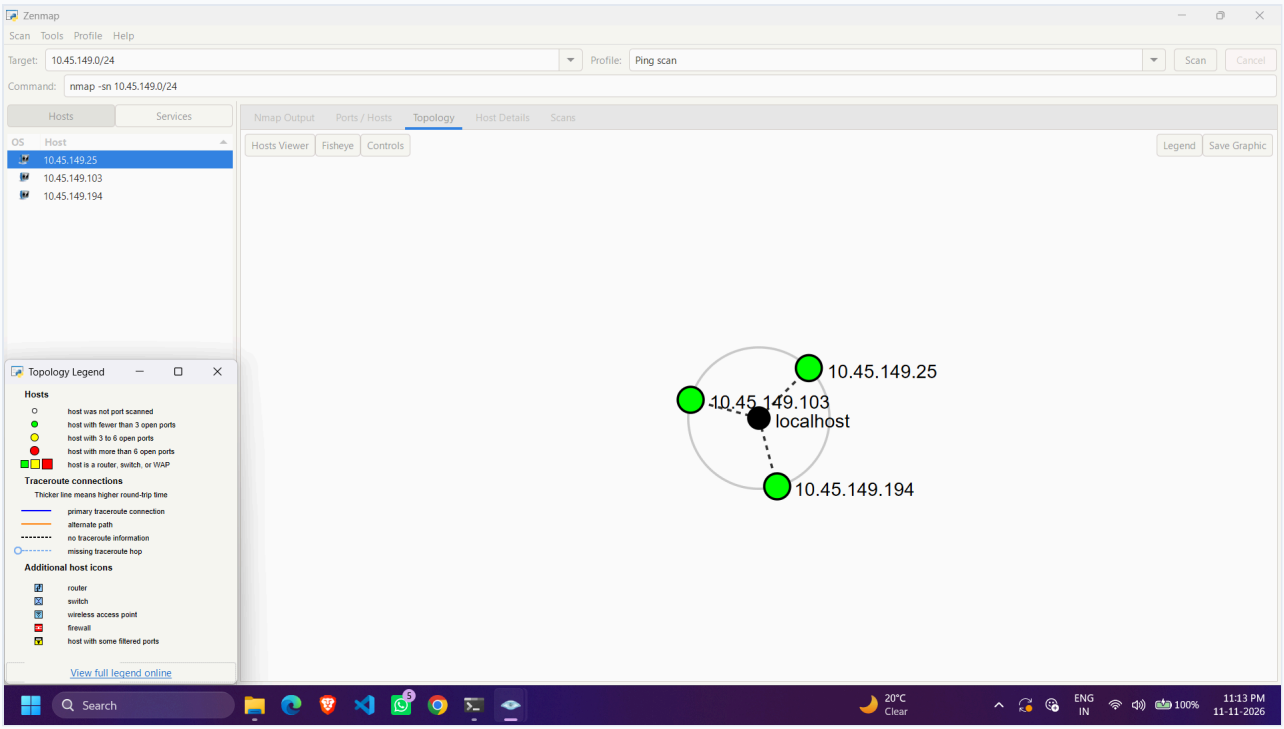


Figure 11: Zenmap Topology View with Legend. Radial topology diagram in Zenmap displaying `localhost` linked to `10.45.149.25`, `10.45.149.103`, and `10.45.149.194` alongside the Topology Legend (exported to `W2-PM5_Network_Topology.pdf`).

Author & Project Information

Author: Naman Sharma | **Contact:** myarcade.era@gmail.com | **Program / Batch:** Cybersecurity Internship at Networkwalks — Batch B084 | **Week / Module:** Week 02 (W2-PM-FINAL)

LinkedIn Profile: www.linkedin.com/in/nonchalant-scholar

GitHub Repository: https://github.com/nonchalant-scholar/NETWORKWALKS-B084-WK2-FOOTPRINTING-AND-NETWORK-SCANNING